

Appendix B: Compilation of Catalogue

Appendix B.1 Platform

The rectangular hollow sections (RHS) are necessary for a large portion of the structural integrity of the platform. The catalogue for the RHS can be seen in Figure 68.

RHS

Dimension d x b mm x mm	End colour codes	THICKNESS T mm	PAINTED	GAL	MASS/METRE KG/M	LENGTHS/BUNDLE QTY x M
50 x 20	RED	1.6	✓		1.63	96 x 8
	YELLOW	2.0	✓		1.99	96 x 8
	PINK	2.5	✓		2.42	72 x 8
	BLUE	3.0			2.83	72 x 8
50 x 25	RED	1.6	✓	✓	1.75	96 x 8
	YELLOW	2.0	✓	✓	2.15	96 x 8
	PINK	2.5	✓	✓	2.62	72 x 8
	BLUE	3.0			3.07	60 x 8
65 x 35	YELLOW	2.0	✓	✓	2.93	54 x 8
	PINK	2.5	✓	✓	3.60	54 x 8
	BLUE	3.0	✓	✓	4.25	45 x 8
	GREEN	4.0			5.35	35 x 8
75 x 25	RED	1.6	✓	✓	2.38	65 x 8
	YELLOW	2.0	✓	✓	2.93	65 x 8
	PINK	2.5			3.60	48 x 8
75 x 50	RED	1.6	✓	✓	3.01	42 x 8
	YELLOW	2.0	✓	✓	3.72	42 x 8
	PINK	2.5	✓	✓	4.58	42 x 8
	BLUE	3.0	✓	✓	5.42	35 x 8
75 x 50	GREEN	4.0	✓	✓	6.92	28 x 8
	ORANGE	5.0	✓		8.35	24 x 8
	WHITE	6.0			9.67	20 x 8
76 x 38	PINK	2.5	✓		4.16	45 x 8
	BLUE	3.0	✓		4.90	40 x 8
	GREEN	4.0			6.23	32 x 8
100 x 50	RED	1.6	✓	✓	3.64	32 x 8
	YELLOW	2.0	✓	✓	4.50	32 x 8
	PINK	2.5	✓	✓	5.56	32 x 8
	BLUE	3.0	✓	✓	6.60	32 x 8
	GREEN	4.0	✓	✓	8.49	24 x 8
	ORANGE	5.0	✓	✓	10.30	18 x 8
	WHITE	6.0			12.00	15 x 8
102 x 76	L GREEN	3.5	✓		9.07	20 x 8
	ORANGE	5.0	✓		12.50	16 x 8
	WHITE	6.0			14.70	12 x 8
125 x 75	BLUE	3.0	✓	✓	8.96	24 x 8
	GREEN	4.0	✓		11.60	20 x 8
	ORANGE	5.0	✓		14.20	15 x 8
	WHITE	6.0			16.70	12 x 8
150 x 50	YELLOW	2.0	✓	✓	6.05	21 x 8
	PINK	2.5	✓	✓	7.53	21 x 8
	BLUE	3.0	✓	✓	8.96	21 x 8
	GREEN	4.0	✓	✓	11.60	15 x 8
	ORANGE	5.0	✓		14.20	15 x 8 & 10 x 12
	WHITE	6.0			16.98	15 x 8
150 x 100	GREEN	4.0	✓		14.80	12 x 8 & 9 x 12
	ORANGE	5.0	✓		18.20	12 x 8 & 8 x 12
	WHITE	6.0	✓		21.40	9 x 8 & 6 x 12
	PURPLE	9.0	✓		30.60	6 x 8 & 4 x 12
	YELLOW	10.0			33.40	4 x 12
152 x 76	ORANGE	5.0	✓		16.40	12 x 8 & 12 x 12
	WHITE	6.0			19.40	9 x 8 & 9 x 12
200 x 100	GREEN	4.0	✓		17.90	8 x 8 & 6 x 12
	ORANGE	5.0	✓		22.10	8 x 8 & 6 x 12
	WHITE	6.0	✓		26.20	8 x 8 & 4 x 12
	PURPLE	9.0	✓		37.70	6 x 8 & 4 x 12
	YELLOW	10.0			41.30	2 x 12
250 x 150	ORANGE	5.0	✓		29.90	6 x 8 & 4 x 12
	WHITE	6.0	✓		35.60	4 x 8 & 4 x 12
	PURPLE	9.0	✓		51.80	4 x 8 & 2 x 12
	YELLOW	10.0	✓		57.00	2 x 12
	BLUE	12.5	✓		69.40	2 x 12
	GREY	16.0			85.50	1 x 12

Figure 68: Rectangular Hollow Section Catalogue. [6]

Aluminium round bars are used as shafts throughout the platform (not including the shaft connected to the motor, which will need to be machined from a billet). The catalogue for the round bars can be seen in Figure 69.

Description	Stock finishes 	Length	Min Sale	Noble Park	Carrum Downs
10mm	MF	4m	1/2	✓	
12mm	MF	4m	1/2	✓	
16mm	MF	4m	1/2	✓	✓
20mm	MF	4m	1/4	✓	✓
25mm	MF	4m	1/4	✓	✓
30mm	MF	4m	1/8	✓	✓
33mm	MF	4m	1/8	✓	
39mm	MF	4m	1/8	✓	✓
50mm	MF	4m	1/8	✓	

Figure 69: Aluminium Round Bars Catalogue. [7]

External circlips are used on many of the shafts on the platform. The catalogue for the external circlips can be seen in Figure 70.

Code	Description	Diameter	Thickness	Thread Type	Material / Finish	Compliance
5MEC	5x0.6mm Plain External Circlip	M5	0.6mm	Metric	Plain	DIN 471
6MEC	6x0.7mm Plain External Circlip	M6	0.7mm	Metric	Plain	DIN 471
7MEC	7x0.8mm Plain External Circlip	M7	0.8mm	Metric	Plain	DIN 471
8MEC	8x0.8mm Plain External Circlip	M8	0.8mm	Metric	Plain	DIN 471
9MEC	9x1.0mm Plain External Circlip	M9	1.0mm	Metric	Plain	DIN 471
10MEC	10x1.0mm Plain External Circlip	M10	1.0mm	Metric	Plain	DIN 471
11MEC	11x1.0mm Plain External Circlip	M11	1.0mm	Metric	Plain	DIN 471
12MEC	12x1.0mm Plain External Circlip	M12	1.0mm	Metric	Plain	DIN 471
13MEC	13x1.0mm Plain External Circlip	M13	1.0mm	Metric	Plain	DIN 471
14MEC	14x1.0mm Plain External Circlip	M14	1.0mm	Metric	Plain	DIN 471
15MEC	15x1.0mm Plain External Circlip	M15	1.0mm	Metric	Plain	DIN 471
16MEC	16x1.0mm Plain External Circlip	M16	1.0mm	Metric	Plain	DIN 471
17MEC	17x1.0mm Plain External Circlip	M17	1.0mm	Metric	Plain	DIN 471
18MEC	18x1.2mm Plain External Circlip	M18	1.2mm	Metric	Plain	DIN 471
19MEC	19x1.2mm Plain External Circlip	M19	1.2mm	Metric	Plain	DIN 471

Figure 70: External Circlips DIN471 Catalogue. [8]

SKF deep groove ball bearings are used as supports for a lot of the shafts on the platform. The catalogue for the deep grooved ball bearings can be seen in Figure 71.

Designation	Principal dimensions				Basic load ratings		Speed ratings	
	d [mm]	t ₁	D [mm]	B [mm]	dynamic	static	Reference speed	Limiting speed
					C [kN]	C ₀ [kN]	[r/min]	[r/min]
☆ W 608-2RS1	8		22	7	1.99	0.78		22 000
☆ W 608-2RS1/VP311	8		22	7	1.99	0.78		22 000
☆ W 619/8-2RS1	8		19	6	1.25	0.455		24 000
W 628-2RS1	8		24	8	2.47	1.12		20 000
☆ W 628/8-2RS1	8		16	5	0.715	0.3		26 000
W 638-2RS1	8		28	9	3.97	1.96		19 000
WBB1-8708-2RS1	8		14	4	0.462	0.193		28 000
☆ W 609-2RS1	9		24	7	2.03	0.815		20 000
W 619/9-2RS1	9		20	6	1.95	0.93		22 000
W 628/9-2RS1	9		17	5	0.761	0.335		24 000
W 629-2RS1	9		26	8	3.97	1.96		19 000
W 639-2RS1	9		30	10	4.36	2.32		16 000
☆ W 6000-2RS1	10		26	8	3.97	1.96		19 000
☆ W 6000-2RS1/VP311	10		26	8	3.97	1.96		19 000
W 61700 X-2RS1	10		15	4	0.488	0.22		24 000
☆ W 61800-2RS1	10		19	5	1.48	0.83		22 000
☆ W 61900-2RS1	10		22	6	2.34	1.25		20 000
☆ W 6200-2RS1	10		30	9	4.36	2.32		16 000
☆ W 6200-2RS1/VP311	10		30	9	4.36	2.32		16 000
☆ W 6300-2RS1	10		35	11	7.02	3.4		15 000
W 63800-2RS1	10		19	7	1.48	0.83		22 000
☆ W 6001-2RS1	12		28	8	4.42	2.36		16 000
☆ W 6001-2RS1/VP311	12		28	8	4.42	2.36		16 000
W 61701-2RS1	12		18	4	0.527	0.265		22 000
W 61802-2RS1	15		24	5	1.65	1.08		17 000
☆ W 61902-2RS1	15		28	7	3.71	2.24		16 000
☆ W 6202-2RS1	15		35	11	6.37	3.6		13 000
☆ W 6202-2RS1/VP311	15		35	11	6.37	3.6		13 000
☆ W 6302-2RS1	15		42	13	9.95	5.4		11 000
W 63802-2RS1	15		24	7	1.65	1.08		17 000
☆ W 6003-2RS1	17		35	10	4.94	3.15		13 000
☆ W 6003-2RS1/VP311	17		35	10	4.94	3.15		13 000
W 61703-2RS1	17		23	4	0.559	0.34		17 000
W 61803-2RS1	17		26	5	1.78	1.27		16 000
☆ W 61903-2RS1	17		30	7	3.97	2.55		14 000
☆ W 6203-2RS1	17		40	12	8.06	4.75		12 000
☆ W 6203-2RS1/VP311	17		40	12	8.06	4.75		12 000
☆ W 6303-2RS1	17		47	14	11.7	6.55		10 000
W 63803-2RS1	17		26	7	1.78	1.27		16 000
☆ W 6004-2RS1	20		42	12	8.06	5		11 000
☆ W 6004-2RS1/VP311	20		42	12	8.06	5		11 000
W 61704-2RS1	20		27	4	0.585	0.39		14 000
☆ W 61804-2RS1	20		32	7	3.12	2.08		13 000
☆ W 61904-2RS1	20		37	9	5.53	3.65		12 000
☆ W 6204-2RS1	20		47	14	10.8	6.55		10 000
☆ W 6204-2RS1/VP311	20		47	14	10.8	6.55		10 000
☆ W 6304-2RS1	20		52	15	13.8	7.8		9 500
W 63804-2RS1	20		32	10	3.12	2.08		13 000

Figure 71: SKF Deep Groove Ball Bearings Catalogue. [9]

Compression springs are used to help fix the position of the guide wheels. Figure 72 illustrates just a snippet of the catalogue for compression springs.

Catalogue ID	Outside Diameter		Wire Diameter		Free Length, L		Load, P		Length, L1		Solid Height Approx.		Spring Rate, R	
	in	mm	in	mm	in	mm	lb	N	in	mm	in	mm	lb/in	N/mm
JWC42	0.375	9.530	0.063	1.600	1.500	38.100	7.600	33.826	1.372	34.860	0.980	24.900	59.610	10.440
C0420-035-0750	0.420	10.670	0.035	0.900	0.750	19.050	4.950	22.020	0.354	8.990	0.211	5.360	12.510	2.190
C0420-035-1000	0.420	10.670	0.035	0.900	1.000	25.400	4.950	22.020	0.458	11.630	0.250	6.350	9.100	1.594
C0420-035-1500	0.420	10.670	0.035	0.900	1.500	38.100	4.950	22.020	0.667	16.940	0.328	8.330	5.900	1.033
C0420-035-1500SS [G304 Stainless]	0.420	10.670	0.035	0.900	1.500	38.100	4.120	18.329	0.667	16.940	0.354	9.000	4.950	0.866
C0420-039-0750	0.420	10.670	0.039	1.000	0.750	19.050	7.210	32.080	0.345	8.760	0.248	6.300	17.760	3.110
C0420-039-1000	0.420	10.670	0.039	1.000	1.000	25.400	7.210	32.080	0.445	11.300	0.310	7.870	12.900	2.260
C0420-039-1500	0.420	10.670	0.039	1.000	1.500	38.100	7.210	32.080	0.650	16.510	0.429	10.900	8.510	1.490
C0420-039-1500SS [G304 Stainless]	0.420	10.670	0.039	1.000	1.500	38.100	5.970	26.552	0.650	16.510	0.423	10.750	7.020	1.230
C0420-039-2000	0.420	10.670	0.039	1.000	2.000	50.800	7.210	32.080	0.976	24.790	0.579	14.710	6.510	1.140
C0420-039-2000SS [G304 Stainless]	0.420	10.670	0.039	1.000	2.000	50.800	5.810	25.824	0.976	24.790	0.496	12.600	5.670	0.993
C0420-039-2500	0.420	10.670	0.039	1.000	2.500	63.500	7.210	32.080	1.209	30.710	0.630	16.000	5.200	0.910
C0420-049-1000	0.420	10.670	0.049	1.250	1.000	25.400	11.970	53.250	0.558	14.170	0.407	10.340	27.120	4.750
C0420-049-1000SS [G304 Stainless]	0.420	10.670	0.049	1.250	1.000	25.400	10.420	46.341	0.558	14.170	0.448	11.375	23.560	4.127
C0420-049-1500	0.420	10.670	0.049	1.250	1.500	38.100	11.970	53.250	0.813	20.650	0.552	14.020	17.420	3.050
C0420-049-2000	0.420	10.670	0.049	1.250	2.000	50.800	11.970	53.250	1.068	27.130	0.698	17.730	12.900	2.260
C0420-055-1000	0.420	10.670	0.055	1.400	1.000	25.400	18.000	80.070	0.600	15.240	0.454	11.530	45.000	7.880
C0420-055-1000SS [G316 Stainless]	0.420	10.670	0.055	1.400	1.000	25.400	15.670	69.700	0.600	15.240	0.441	11.200	39.170	6.860
C0420-055-1500	0.420	10.670	0.055	1.400	1.500	38.100	18.000	80.070	0.880	22.350	0.645	16.380	29.010	5.080
C0420-055-2000	0.420	10.670	0.055	1.400	2.000	50.800	18.000	80.070	1.179	29.950	0.865	21.970	21.300	3.730
C0420-055-2500	0.420	10.670	0.055	1.400	2.500	63.500	18.000	80.070	1.430	36.320	1.050	26.670	16.790	2.940
C0420-055-2500SS [G304 Stainless]	0.420	10.670	0.055	1.400	2.500	63.500	15.630	69.518	1.430	36.320	1.061	26.950	14.600	2.558
C0420-063-1000	0.420	10.670	0.063	1.600	1.000	25.400	25.370	112.850	0.675	17.150	0.591	15.010	77.890	13.640
C0420-063-1500	0.420	10.670	0.063	1.600	1.500	38.100	25.370	112.850	0.988	25.100	0.821	20.850	49.620	8.690
C0420-063-2000	0.420	10.670	0.063	1.600	2.000	50.800	25.370	112.850	1.302	33.070	1.052	26.720	36.320	6.360
C0420-063-2500	0.420	10.670	0.063	1.600	2.500	63.500	25.370	112.850	1.616	41.050	1.282	32.560	28.720	5.030
JWC29	0.437	11.110	0.039	1.000	1.500	38.100	4.090	18.173	0.714	18.130	0.510	12.950	5.200	0.910
JWC39	0.437	11.110	0.055	1.400	2.000	50.800	18.060	80.344	1.119	28.420	0.799	20.300	20.500	3.590
JWC43	0.437	11.110	0.063	1.600	2.000	50.800	22.170	98.607	1.328	33.740	0.949	24.100	33.000	5.780
C0480-035-1000	0.480	12.190	0.035	0.900	1.000	25.400	4.200	18.690	0.370	9.400	0.220	5.590	6.680	1.170

Figure 72: Compression Springs Catalogue. [10]

Guide wheels provide the platform stability. The catalogue for guide wheels can be seen in Figure 73.

TYPE	CODE	WHEEL DIAMETER (MM)	TYRE WIDTH	WHEEL TYPE	BEARING TYPE	BOLT HOLE SIZE	CAPACITY	HEIGHT (MM)
								
BOLT HOLE	CGZGRQ75BH	75	23.5	 RUBBER	BALL	10.5	70kg	100
BOLT HOLE / BRAKE	CGZGRQ75BHTB	75	23.5		BALL	10.5	70kg	100
BOLT HOLE	CGZGRQ100BH	100	31		BALL	12.5	100kg	131
BOLT HOLE / BRAKE	CGZGRQ100BHTB	100	31		BALL	12.5	100kg	131
BOLT HOLE	CGZGRQ125BH	125	32		BALL	12.5	120kg	162
BOLT HOLE / BRAKE	CGZGRQ125BHTB	125	32		BALL	12.5	120kg	162
BOLT HOLE	CGZPUQ75BH	75	23.5	 URETHANE	BALL	10.5	70kg	100
BOLT HOLE / BRAKE	CGZPUQ75BHTB	75	23.5		BALL	10.5	70kg	100
BOLT HOLE	CGZPUQ100BH	100	31		BALL	12.5	100kg	131
BOLT HOLE / BRAKE	CGZPUQ100BHTB	100	31		BALL	12.5	100kg	131
BOLT HOLE	CGZPUQ125BH	125	32		BALL	12.5	120kg	162
BOLT HOLE / BRAKE	CGZPUQ125BHTB	125	32		BALL	12.5	120kg	162

Figure 73: Guide Wheels Catalogue. [11]

Aluminium sheet metal is necessary for a few parts in the platform, such as brackets. Catalogue for aluminium sheet metal can be seen in Figure 74.

Part Number	A mm	B mm	Alloy/ Temper
C0.551220MI2TJ	0.55	1220	3105-H26
C0.61200EM2FD	0.60	1200	5005-H34
C0.61200MI2FD	0.60	1200	5005-H34
C0.71200MI2FD	0.70	1200	5005-H34
C0.81200MI2FD	0.80	1200	5005-H34
C0.82515MI2TA	0.80	2515	3003-H16
C0.82515MI2TAM	0.80	2515	3003-H16
C0.91200MI2FD	0.90	1200	5005-H34
C0.91200MI2FW	0.90	1200	5005-H16
C1.01200MI2FD	1.0	1200	5005-H34
C1.01200MI2FI	1.0	1200	5052-H32
C1.21200AG2FD	1.20	1200	5005-H34
C1.21200MI2FI	1.20	1200	5052-H32
C1.21200MI2FI	1.20	1200	5052-H32
C1.61200MI2FD	1.60	1200	5005-H34
C1.61200MI2FI	1.60	1200	5052-H32
C2.01200MI2FD	2.0	1200	5005-H34
C2.01200MI2FI	2.0	1200	5052-H32
C3.01200MI2FI	3.0	1200	5052-H32
C3.01200PV1FD	3.0	1200	5005-H34
C3.01500PV1FD	3.0	1500	5005-H34

Figure 74: Aluminium Sheet Metal Catalogue. [12]

Aluminium propeller treadplate sheets are used on the base on the platform for strength and traction. Figure 75 illustrates the catalogue for Aluminium propeller treadplate sheets.

Part Number	T mm	W mm	L mm	Alloy/Temper	Finish	Nominal Mass/Piece
TP1.612002400PP2TI	1.6	1200	2400	3003-H224	Mill	14.423
TP1.612003000PP2TI	1.6	1200	3000	3003-H224	Mill	18.028
TP1.612003600PP2TI	1.6	1200	3600	3003-H224	Mill	21.634
TP2.012002400PP2TI	2.0	1200	2400	3003-H224	Mill	17.28
TP3.012002400PP2TI	3.0	1200	2400	3003-H224	Mill	25.660
TP5.09003600PP2TI	5.0	900	3600	3003-H224	Mill	47.952
TP5.012002400PP2TI	5.0	1200	2400	3003-H224	Mill	42.624

Figure 75: Aluminium Propeller Treadplate Sheet Catalogue. [13]

Hinges and a latch are used to access and use the storage compartment at the bottom of the platform. A catalogue for hinges and latches can be seen in Figure 76 and Figure 77, respectively.

Part number	Material	Finish	A	B	C	D	E	F	G	H	S	Weight (g)
72-7-3662	brass	raw	40	30	2	3	2.5	18	14		3	20.5
72-7-3663	brass	raw	40	35	2	3	2.5	21	14		3	24
72-7-3664	brass	raw	40	40	2	3	2.5	23	14		3	29
72-7-3665	brass	raw	50	30	2	3	2.5	17.5	17.5		3	26
72-7-3666	brass	raw	60	35	2	3	2.5	20	21.5	24	3.5	37
72-7-3667	brass	raw	60	40	2	3	2.75	21.5	21.5	26	3.5	43.5

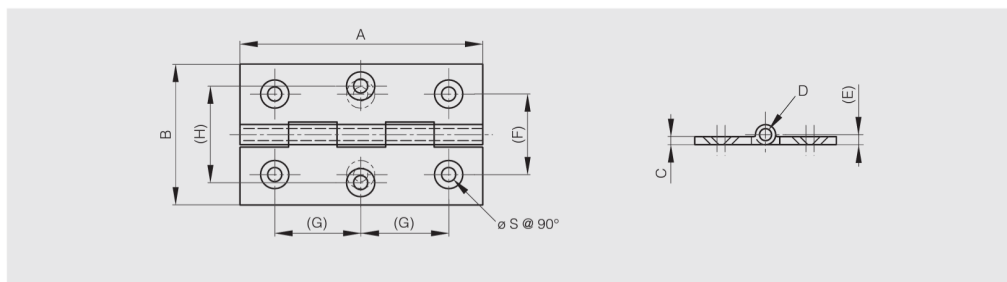


Figure 76: Hinge Catalogue. [14]

Part number	Material	Finish	A	B	Weight (g)
16-7-3680	brass	polished	40	30	66
16-7-3681	brass	chrome plated	40	30	66

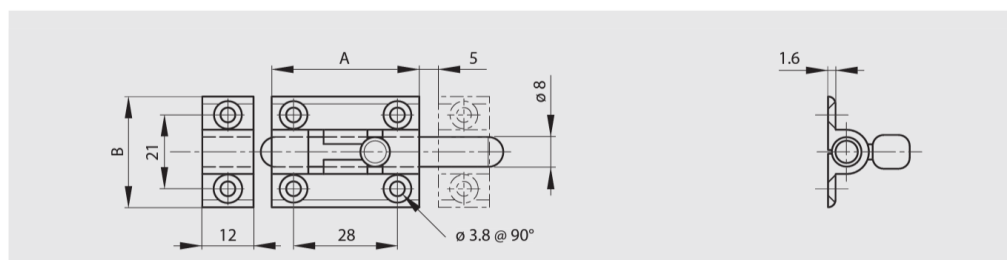


Figure 77: Latch Catalogue. [14]

	Aluminium Rivet Dome Head Aluminium Head/Steel Mandrel	
	* - Stainless Steel Head/Stainless Steel Mandrel * - Aluminium Head/Aluminium Mandrel	
Code	Diameter	Length
AS3-02	2.4mm (3/32")	6.0mm
AS3-03	2.4mm (3/32")	8.0mm
AS3-04	2.4mm (3/32")	9.0mm
AS4-01 *	3.2mm (1/8")	4.5mm
AS4-02 **	3.2mm (1/8")	6.2mm
AS4-03 **	3.2mm (1/8")	7.8mm
AS4-04 **	3.2mm (1/8")	9.4mm
AS4-05 **	3.2mm (1/8")	10.9mm
AS4-06 **	3.2mm (1/8")	12.5mm
AS4-08 **	3.2mm (1/8")	16.0mm
AS4-10	3.2mm (1/8")	19.0mm
AS4-12	3.2mm (1/8")	23.5mm
AS5-01 *	4.0mm (5/32")	5.5mm
AS5-02 **	4.0mm (5/32")	6.8mm
AS5-03 **	4.0mm (5/32")	8.4mm
AS5-04 **	4.0mm (5/32")	10.0mm
AS5-05 **	4.0mm (5/32")	11.9mm
AS5-06 **	4.0mm (5/32")	13.0mm
AS5-08 **	4.0mm (5/32")	16.5mm
AS5-10 *	4.0mm (5/32")	20.0mm
AS5-12	4.0mm (5/32")	24.0mm
AS6-01	4.8mm (3/16")	6.5mm
AS6-02 **	4.8mm (3/16")	7.5mm
AS6-03 *	4.8mm (3/16")	9.0mm
AS6-04 **	4.8mm (3/16")	10.5mm
AS6-05 *	4.8mm (3/16")	12.7mm
AS6-06 **	4.8mm (3/16")	13.8mm
AS6-08 **	4.8mm (3/16")	17.5mm
AS6-10 **	4.8mm (3/16")	20.5mm
AS6-12 **	4.8mm (3/16")	23.5mm
AS6-14 *	4.8mm (3/16")	27.0mm
AS6-16 *	4.8mm (3/16")	30.0mm
AS6-20	4.8mm (3/16")	35.0mm
AS6-22	4.8mm (3/16")	40mm
AS8-02	6.4mm (1/4")	8.7mm
AS8-04 *	6.4mm (1/4")	11.9mm
AS8-05	6.4mm (1/4")	14.0mm
AS8-06 *	6.4mm (1/4")	15.0mm
AS8-08 *	6.4mm (1/4")	18.5mm
AS8-10 *	6.4mm (1/4")	21.9mm
AS8-12 *	6.4mm (1/4")	25.0mm
AS8-14	6.4mm (1/4")	28.3mm
AS8-16	6.4mm (1/4")	31.4mm
	Self Tapping Screws Pan Head Stainless Steel Available in Phillips & Square drive	
Gauge	Length in mm	
4g	6, 10, 12, 16, 20, 25	
6g	10, 12, 16, 20, 25, 32, 38, 50	
8g	10, 12, 16, 20, 25, 32, 38, 45, 50	
10g	12, 16, 20, 25, 32, 38, 45, 50	
12g	12, 20, 25, 32, 38, 50	
	Metric Bolt Available in Plain, Zinc & *Stainless Steel (A2-70)	
	Class 8.8	
Diameter	Min. Length	Max. Length
M4 *	8mm	40mm
M5 *	8mm	80mm
M6 *	10mm	130mm
M8 *	10mm	200mm
M10 *	12mm	220mm
M12 *	20mm	220mm
M14 *	25mm	200mm
M16 *	25mm	300mm
M18 *	30mm	100mm
M20 *	30mm	300mm
M22 *	55mm	400mm
M24 *	40mm	300mm
M27 *	70mm	400mm
M30 *	75mm	450mm
M36 *	90mm	300mm

While purchasing electronics may be out of a scope of this mechanical project, a relevant catalogue where one can find buttons, dials, E-stops and more can be seen in Figure 79.



General Purpose
Cat. No. 800Z-GF2Q5



General Purpose
Cat. No. 800Z-GL3Q5B

Mounting Hole Size	Input Voltage	Output Type	Electrical Connection	No Guard	Black Guard
				Cat. No.	Cat. No.
30.5 mm	100...240V AC	Relay Output	5-Pin QD	800Z-GF3Q5	800Z-GF3Q5B
			6 ft Cabled — 5-Wire	800Z-GF3Q65	800Z-GF3Q65B
	10...40V DC and 20...30V AC	Relay Output	5-Pin QD	800Z-GL3Q5	800Z-GL3Q5B
			6 ft Cabled — 5-Wire	800Z-GL3Q65	800Z-GL3Q65B
22.5 mm	100...240V AC	Relay Output	5-Pin QD	800Z-GF2Q5	800Z-GF2Q5B
			6 ft Cabled — 5-Wire	800Z-GF2Q65	800Z-GF2Q65B
	10...40V DC and 20...30V AC	Relay Output	5-Pin QD	800Z-GL2Q5	800Z-GL2Q5B
			6 ft Cabled — 5-Wire	800Z-GL2Q65	800Z-GL2Q65B

Booted Knobs



Booted Standard Knob Selector Switch
Cat. No. 800H-NP5



Booted Knob Lever Selector Switch
Cat. No. 800H-NPE5

Switch Type	Boot Material*			
	Chlorosulfonated Polyethylene	Silicone *	Urethane **	Ethylene * Propylene **
	Cat. No.	Cat. No.	Cat. No.	Cat. No.
Standard Knob Selector Switch*	800H-NP5	800H-NP101	800H-NP100	800H-NP103
Knob Lever Selector Switch*	800H-NPE5	800H-NPE101	800H-NPE100	800H-NPE103

* Dots printed on the inside of the boot identify the boot material.

* Includes color insert kit.

Note: A jumbo or large legend plate is recommended, if space allows.



2-Position Push-Pull
Cat. No. 800T-FX6D4



2-Position Metal Push-Pull
Cat. No. 800T-FXLE6D4S



2-Position Push-Pull / Twist
Cat. No. 800T-FXT6D4



2-Position Push-Pull / Twist
Cat. No. 800H-FRXT6D4

Contact Type	Operator Position		Type 4/13		Type 4/4X/13	
	Out	In	Push-Pull — 40 mm	Push-Pull — 63 mm Metal	Push-Pull/Twist Release	Push-Pull/Twist Release
			Cat. No.	Cat. No.	Cat. No.	Cat. No.
N.C.L.B.*	X	O	800T-FX6D4	800T-FXLE6D4	800T-FXT6D4	800H-FRXT6D4
N.O. - N.C.L.B.*	O	X	800T-FX6A1	800T-FXLE6A1	800T-FXT6A1	800H-FRXT6A1
N.C.L.B. - N.C.L.B.*	X	O	800T-FX6A5	800T-FXLE6A5	800T-FXT6A5	800H-FRXT6A5
S.M.C.B.*	X	O	800TC-FX6D4S	800TC-FXLE6D4S	800TC-FXT6D4S	800HC-FRXT6D4S
N.O. - S.M.C.B.*	O	X	800TC-FX6A1S	800TC-FXLE6A1S	800TC-FXT6A1S	800HC-FRXT6A1S
S.M.C.B. - S.M.C.B.*	X	O	800TC-FX6A5S	800TC-FXLE6A5S	800TC-FXT6A5S	800HC-FRXT6A5S

Figure 79: Electronic Controls Catalogue. [16]

Pillow Block bearing assembly catalogue for SKF selections, necessary for shaft attachment and stabilisation, can be seen in Figure 80.

Designation	Dimensions						Basic load ratings		Limiting speed	Included products	
							dynamic	static		Housing	Bearing
	d [mm]	A [mm]	H [mm]	H ₂ [mm]	J [mm]	L [mm]	C [kN]	T ₁	C ₀ [kN]	[r/min]	
P 20 FM	20	32	25.3	50.5	76	98.5	12.7	6.55	8 500	P 47	YET 204
P 20 RM	20	32	25.3	50.5	76	98.5	12.7	6.55	8 500	P 47	YAT 204
P 20 TF	20	32	25.3	50.5	76	98.5	12.7	6.55	8 500	P 47	YAR 204-2F
P 20 TR	20	32	25.3	50.5	76	98.5	12.7	6.55	5 000	P 47	YAR 204-2RF
P 20 WF	20	32	25.3	50.5	76	98.5	12.7	6.55	8 500	P 47	YEL 204-2F
P 52 R-20 FM	20	32	28.4	56	86	108	12.7	6.55	8 500	P 52	YET 204
P 52 R-20 RM	20	32	28.4	56	86	108	12.7	6.55	8 500	P 52	YAT 204
P 52 R-20 TF	20	32	28.4	56	86	108	12.7	6.55	8 500	P 52	YAR 204-2F
P 52 R-20 WF	20	32	28.4	56	86	108	12.7	6.55	8 500	P 52	YEL 204-2F
P2B 20M-TF	20	34	33.34	64.29	95.25	127	12.7	6.55	8 500	P2B 504	YAR 204-2F
P2BC 20M-TPZM	20	36	33.3	64.6	95	129	12.7	6.55	5 000	P2BC 504	YAR 204-2LPW/ZM
SY 20 FM	20	32	33.3	64.5	97	127	12.7	6.55	8 500	SY 504 M	YET 204
SY 20 TDW	20	32	33.3	64.5	97	127	12.7	6.55	1 800	SY 504 M/VZ033	YAR 204-2DW/AG
SY 20 TF	20	32	33.3	64.5	97	127	12.7	6.55	8 500	SY 504 M	YAR 204-2F
SY 20 TR	20	32	33.3	64.5	97	127	12.7	6.55	5 000	SY 504 M	YAR 204-2RF
SY 20 WDW	20	32	33.3	64.5	97	127	12.7	6.55	1 800	SY 504 M/VZ033	YEL 204-2DW/AG
SY 20 WF	20	32	33.3	64.5	97	127	12.7	6.55	8 500	SY 504 M	YEL 204-2F
SVJ 20 TF	20	34	33.3	64.5	97	127	12.7	6.55	8 500	SVJ 504	YAR 204-2F
SYK 20 TD	20	32	33.3	64.5	96.5	126	12.7	6.55	1 800	SYK 504	YAR 204-2DW/AG
SYK 20 TF	20	32	33.3	64.5	96.5	126	12.7	6.55	8 500	SYK 504	YAR 204-2F
SYK 20 TR	20	32	33.3	64.5	96.5	126	12.7	6.55	5 000	SYK 504	YAR 204-2RF
SYK 20 WD	20	32	33.3	64.5	96.5	126	12.7	6.55	1 800	SYK 504	YEL 204-2DW/AG
SYWK 20 YTA	20	32	33.3	64	96.5	126	12.7	6.55	5 000	SYWK 504 Y	YAR 204-2RF/VE495

Figure 80: Pillow Block Bearing Assemblies Catalogue. [17]

4140 Steel Round Bar used as the block for shaft and pinion milling can be seen in Figure 81. This alloy steel is chosen due to its high yield and tensile strengths.

COMPLETE STEEL & METAL SOLUTIONS Search Q 1300 233 266 ALEX MAVRIDS MAKE PAYMENT 0 ITEM(S)					
SHOP ONLINE Steel Metal Livestock & Fencing Tools & Gear SPECIALS					
STEEL > 4140 STEEL > ROUND BAR > 4140					
STOCK CODE	DESCRIPTION	DIMENSIONS (MM)	STANDARD LENGTH (MM)	KG PER UNIT	
4140RND.28	28.0mm 4140/709M Peeled Round bar	28	6000	4.82kg	VIEW PRICE >
4140RND.30	30.0mm 4140/709M Peeled Round bar	30	6000	5.55kg	VIEW PRICE >
4140RND.35	35.0mm 4140/709M Peeled Round Bar	35	6000	7.55kg	VIEW PRICE >
4140RND.36	36.0mm 4140/709M Peeled Round bar	36	6000	7.99kg	VIEW PRICE >
4140RND.40	40.0mm 4140/709M Peeled Round bar	40	6000	9.87kg	VIEW PRICE >
4140RND.45	45.0mm 4140/709M Peeled Round bar	45	6000	12.49kg	VIEW PRICE >
4140RND.50	50.0mm 4140/709M Peeled Round bar	50	6000	15.41kg	VIEW PRICE >
4140RND.55	55.0mm 4140/709M Peeled Round bar	55	6000	18.65kg	VIEW PRICE >
4140RND.60	60.0mm 4140/709M Peeled Round bar	60	6000	22.20kg	VIEW PRICE >
4140RND.65	65.0mm 4140/709M Peeled Round bar	65	6000	26.05kg	VIEW PRICE >
4140RND.70	70.0mm 4140/709M Peeled Round bar	70	6000	30.21kg	VIEW PRICE >
4140RND.75	75.0mm 4140/709M Peeled Round bar	75	6000	34.68kg	VIEW PRICE >
4140RND.82	82.0mm 4140/709M Peeled Round bar	82	6000	41.42kg	VIEW PRICE >
4140RND.85	85.0mm 4140/709M Peeled Round bar	85	6000	46.63kg	VIEW PRICE >
4140RND.90	90.0mm 4140/709M Peeled Round bar	90	6000	52.14kg	VIEW PRICE >
4140RND.95	95.0mm 4140/709M Peeled Round bar	95	6000	57.96kg	VIEW PRICE >
4140RND.100	100.00mm 4140/709M Peeled Round Bar	100	6000	64.09kg	VIEW PRICE >

Figure 81: 4140 Steel Round Bar Catalogue [18]

Siemens Simogear catalogue from which the motors were selected, can be found in Figure 82. Chosen due to the correct torque and speed outputs required for each shaft.

SIMOGEAR geared motors							
Helical worm geared motors							
Geared motors up to 7.5 kW							
Selection and ordering data (continued)							
P_{rated} kW	n_2 rpm	T_2 Nm	i -	F_{R2} N	f_B -	m kg	Article No. (Article No. supplement → below) Order code No. of poles
0.55	C.49-LA71ZML4						
	13	295	105.00	7 110	1.2	22	2KJ3603 - ■ CH11 - ■ ■ D2
	15	260	91.93	6 950	1.4	22	2KJ3603 - ■ CH11 - ■ ■ C2
	17	225	80.60	6 810	1.5	22	2KJ3603 - ■ CH11 - ■ ■ B2
	19	205	73.12	6 680	1.5	22	2KJ3603 - ■ CH11 - ■ ■ A2
	20	195	68.82	6 590	1.6	22	2KJ3603 - ■ CH11 - ■ ■ X1
	23	172	60.67	6 420	1.7	22	2KJ3603 - ■ CH11 - ■ ■ W1
	26	149	52.65	6 220	1.8	22	2KJ3603 - ■ CH11 - ■ ■ V1
	27	170	49.87	5 670	1.9	22	2KJ3603 - ■ CH11 - ■ ■ U1
	32	148	43.27	5 530	2.4	22	2KJ3603 - ■ CH11 - ■ ■ T1
	C.39-LE80MB4						
	16	240	91.93	6 000	0.80	19	2KJ3602 - ■ DB21 - ■ ■ C2
	18	210	80.60	6 110	0.91	19	2KJ3602 - ■ DB21 - ■ ■ B2
	20	194	73.12	6 170	1.0	19	2KJ3602 - ■ DB21 - ■ ■ A2
	21	183	68.82	6 210	1.1	19	2KJ3602 - ■ DB21 - ■ ■ X1
	24	161	60.67	6 300	1.1	19	2KJ3602 - ■ DB21 - ■ ■ W1
	27	140	52.65	6 370	1.2	19	2KJ3602 - ■ DB21 - ■ ■ V1
	29	162	49.87	6 290	1.2	19	2KJ3602 - ■ DB21 - ■ ■ U1
	33	140	43.27	6 370	1.4	19	2KJ3602 - ■ DB21 - ■ ■ T1
	37	127	39.33	6 420	1.6	19	2KJ3602 - ■ DB21 - ■ ■ S1
	43	109	33.73	6 490	1.8	19	2KJ3602 - ■ DB21 - ■ ■ R1
	44	108	32.64	6 490	2.0	19	2KJ3602 - ■ DB21 - ■ ■ Q1
	51	94	28.32	6 290	2.5	19	2KJ3602 - ■ DB21 - ■ ■ P1
	56	86	25.75	6 150	2.7	19	2KJ3602 - ■ DB21 - ■ ■ N1
	C.39-LA71ZML4						
	17	220	80.60	6 080	0.87	16	2KJ3602 - ■ CH11 - ■ ■ B2
	19	200	73.12	6 150	0.95	16	2KJ3602 - ■ CH11 - ■ ■ A2
	20	192	68.82	6 180	1.0	16	2KJ3602 - ■ CH11 - ■ ■ X1
	23	169	60.67	6 270	1.1	16	2KJ3602 - ■ CH11 - ■ ■ W1
	26	147	52.65	6 350	1.2	16	2KJ3602 - ■ CH11 - ■ ■ V1
	27	170	49.87	6 260	1.2	16	2KJ3602 - ■ CH11 - ■ ■ U1
	32	147	43.27	6 350	1.4	16	2KJ3602 - ■ CH11 - ■ ■ T1
	35	134	39.33	6 400	1.5	16	2KJ3602 - ■ CH11 - ■ ■ S1
	41	115	33.73	6 470	1.7	16	2KJ3602 - ■ CH11 - ■ ■ R1
	42	114	32.64	6 470	1.9	16	2KJ3602 - ■ CH11 - ■ ■ Q1
	48	99	28.32	6 360	2.4	16	2KJ3602 - ■ CH11 - ■ ■ P1
	53	90	25.75	6 230	2.6	16	2KJ3602 - ■ CH11 - ■ ■ N1
	62	77	22.08	6 010	3.0	16	2KJ3602 - ■ CH11 - ■ ■ M1
	C.29-LE80MB4						
	37	127	39.33	4 040	0.81	13	2KJ3601 - ■ DB21 - ■ ■ S1
	43	110	33.73	4 130	0.95	13	2KJ3601 - ■ DB21 - ■ ■ R1
	44	107	32.64	4 140	0.84	13	2KJ3601 - ■ DB21 - ■ ■ Q1
	51	93	28.32	4 210	0.96	13	2KJ3601 - ■ DB21 - ■ ■ P1
	56	85	25.75	4 250	1.1	13	2KJ3601 - ■ DB21 - ■ ■ N1
	65	73	22.08	4 310	1.2	13	2KJ3601 - ■ DB21 - ■ ■ M1
	72	67	20.07	4 350	1.4	13	2KJ3601 - ■ DB21 - ■ ■ L1
	82	59	17.60	4 390	1.6	13	2KJ3601 - ■ DB21 - ■ ■ K1
	92	52	15.71	4 370	1.8	13	2KJ3601 - ■ DB21 - ■ ■ J1
	103	47	14.00	4 250	2.0	13	2KJ3601 - ■ DB21 - ■ ■ H1

Figure 82: Siemens SIMOGEAR Motor Catalogue [19]

Appendix B.2 Safety Brace

In order to machine the parts for the safety brace mechanism a sheet of alloy steel is needed to be purchased, Figure 83 shows the price of steel per kg.

Alloy Steel Price Per Kg

Shape wise Approx Alloy Price Per Kg	Price in INR	Price in USD
ALLOY STEEL SEAMLESS PIPES GRADE ASTMA335 (SIZE 323.8X12.7 MM)	INR 130	USD 1.79
ALLOY STEEL SEAMLESS PIPE-SPEC- SA335 SIZE:OD 339.7MM X WT:10MM X L:5350MM	INR 2,699	USD 37.25
ALLOY STEEL SEAMLESS STEEL TUBES O.D. 51.55MM X W/T 6.35MM	INR 200	USD 2.76
ALLOY STEEL PLATES GRADE S50C+CR, SIZE THIK 55-90MMXWID	INR 3016	USD 41.63
PIPE FITTINGS ALLOY STEEL : DN400, 90 DEGREE ELBOW (LR), SCH60, SA182 GR.WP5	INR 186	USD 2.57
FLANGES ALLOY STEEL SA182 Pg1 DN350 THK 49.53MM	INR 1,826	USD 25.20

Figure 83: Alloy Steel Plate per kg [20]

In order to attach the person to the safety brace mechanism, a snap hook (carabiner) is needed, available snap hooks to purchase are shown in Figure 84.

CODE	CHAIN SIZE (mm)	B (mm)	E (mm)	H (mm)
520304	4	7	7	40
520305	5	8	7	50
520306	6	9	8	60
520308	8	12	9	80
520310	10	15	11	100
520312	12	20	19	140
520315	15	22	35	200

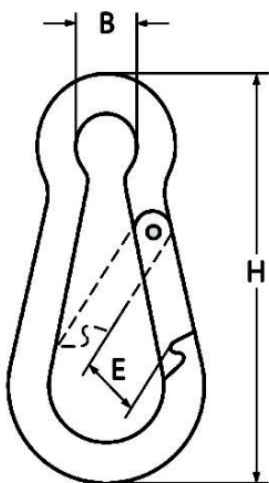
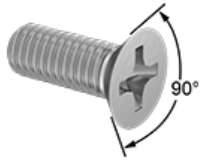


Figure 84: Snap Hook Catalogue. [21]

To join both of the side pieces of the safety brace together, 4 M4x0.7 metric screws are needed, available M4x0.7 metric screws are shown in Figure 85.

Metric 316 Stainless Steel Phillips Flat Head Screws



More corrosion resistant than 18-8 stainless steel screws, these screws have excellent resistance to chemicals and salt water. They're passivated for added protection against oxidation and corrosion. Screws are angled under the head to sit flush within countersunk holes. They may be mildly magnetic. Length is measured from the top of the head.

 For technical drawings and 3-D models, click on a part number.

Lg., mm	Threading	Head Dia., mm	Head Ht., mm	Drive Size	Tensile Strength, psi	Specifications Met	Pkg. Qty.	Pkg.	
316 Stainless Steel									
M4 × 0.7 mm									
18	Fully Threaded	7.5	2.2	No. 2	70,000	DIN 965	50	91801A670	\$14.00
20	Fully Threaded	7.5	2.2	No. 2	70,000	DIN 965	50	91801A224	12.87
22	Fully Threaded	7.5	2.2	No. 2	70,000	DIN 965	50	91801A914	12.80
25	Fully Threaded	7.5	2.2	No. 2	70,000	DIN 965	25	91801A227	11.94

Figure 85: M4x0.7 Screws Catalogue [22]

To make the push button of the safety brace, push back into a closed position (closed meaning that the mechanism cannot disconnect from the cable) a 15 mm spring is needed, available springs are shown in Figure 86.

Corrosion-Resistant Compression Springs



These springs are more corrosion resistant than standard compression springs. They're also easier to compress than standard compression springs. As you squeeze a compression spring, it pushes back to return to its original length. Spring rate is the amount of force required for every inch of compression or, for metric springs, millimeter of compression. The higher the spring rate, the harder it is to compress the spring.

302 stainless steel springs have good corrosion resistance.

316 stainless steel springs have excellent corrosion resistance.

Springs with closed and ground ends sit flat, so they won't buckle.

 For technical drawings and 3-D models, click on a part number.

Metric										
OD, mm	ID, mm	Wire Dia., mm	Compressed Lg. @ Max. Load, mm	Max. Load , lbs.	Spring Rate, lbs./mm	Material	End Type	Specifications Met	Pkg. Qty.	Pkg.
15 mm Lg.										
9.25	6.75	1.25	7.4	19.65	2.67	302 Stainless Steel	Closed and Ground	DIN 2095	5	2006N236 \$15.67
9.25	6.75	1.25	11.47	9.46	2.68	Passivated 316 Stainless Steel	Closed and Ground	—	1	8969T856 4.27

Figure 86: 15 mm Compression Spring Catalogue [23]

For connection of the safety brace to the wind tunnel structure, a steel cable is needed, available steel cables are shown in Figure 87.

Diameter	Breaking Strength kgs	Stretch at BLL %	Minimum Sheave Diameter mm	Construction	Weight kgs per metre
3mm	499	1.28%	72	6X19 fiber core	0.03
6mm	1995	1.28%	144	6X19 fiber core	0.13
8mm	3547	1.28%	192	6X19 fiber core	0.22
10mm	5560	1.28%	240	6X19 fiber core	0.35
12mm	7995	1.28%	288	6X19 fiber core	0.51

Figure 87: Steel Cable Catalogue [24]

Appendix B.3 Scissor lift

The catalogue for the scissor lift can be seen in Figure 88.

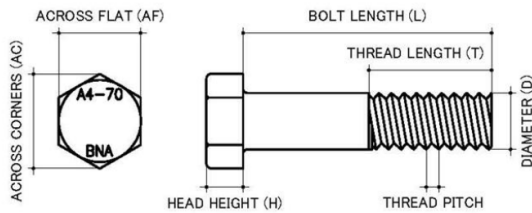
Model	Load Capacity (KG)	Self Height (MM)	Max Platform Height(MM)	Platform Size(MM) LxW	Base Size (MM) LxW	Lifting time (S)	Net weight (KG)
HW1001	1000	205	1000	1300x820	1240x640	20~25	160
HW1002	1000	205	1000	1600x1000	1240x640	20~25	186
HW1003	1000	240	1300	1700x850	1580x640	30~35	200
HW1004	1000	240	1300	1700x1000	1580x640	30~35	210
HW1005	1000	240	1300	2000x850	1580x640	30~35	212
HW1006	1000	240	1300	2000x1000	1580x640	30~35	223
HW1007	1000	240	1300	1700x1500	1580x1320	30~35	365
HW1008	1000	240	1300	2000x1700	1580x1320	30~35	430
HW2001	2000	230	1000	1300x850	1220x785	20~25	235
HW2002	2000	230	1050	1600x1000	1280x785	20~25	268
HW2003	2000	250	1300	1700x850	1600x785	25~35	289
HW2004	2000	250	1300	1700x1000	1600x785	25~35	300
HW2005	2000	250	1300	2000x850	1600x785	25~35	300
HW2006	2000	250	1300	2000x1000	1600x785	25~35	315
HW2007	2000	250	1400	1700x1500	1600x1435	25~35	415
HW2008	2000	250	1400	2000x1800	1600x1435	25~35	500
HW4001	4000	240	1050	1700x1200	1600x900	30~40	375
HW4002	4000	240	1050	2000x1200	1600x900	30~40	405
HW4003	4000	300	1400	2000x1000	1980x900	35~40	470
HW4004	4000	300	1400	2000x1200	1980x900	35~40	490
HW4005	4000	300	1400	2200x1000	2000x900	35~40	480
HW4006	4000	300	1400	2200x1200	2000x900	35~40	505
HW4007	4000	350	1300	1700x1500	1620x1400	35~40	570
HW4008	4000	350	1300	2200x1800	1620x1400	35~40	655
HD1000	1000	1780	305	1300x820	1240x640	35-45	210
HD2000	2000	1780	350	1300x820	1220x785	35-45	295
HD4000	4000	2050	400	1700x1200	1600x900	55-65	520
HT1000	1000	3000	470	1700x1000	1600x1000	35-45	450
HT2000	2000	3000	560	1700x1000	1600x1100	50-60	750
HF400	400	4140	600	1700x1000	1600x1000	30-40	800
HF800	800	4210	706	1700x1000	1606x1010	70-80	858

Figure 88: Scissor Lift Catalogue [3]

Appendix B.4 Tower Clamping System

To connect the channel beams to the tower wall, we use the fasteners seen in Figure 89..

Technical Specifications

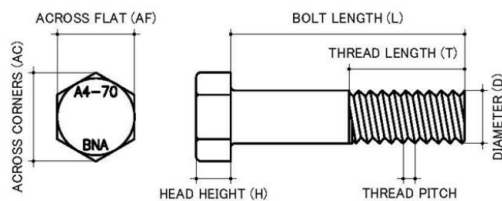


QFind	: 51S10M
System of Measurement	: Metric
Product	: Bolt Only
Head Type	: Hex Head
Size	: M10 (10mm)
Size / Diameter	: 10mm M10
Size & Thread Pitch	: M10 x 1.50p Coarse
Thread Pitch	: 1.50mm
Thread Direction	: Right Hand
Thread Type	: Metric - Coarse
Class / Grade	: A4-70 G316
Material	: Stainless Steel
Corrosion Protection Level	: Ultimate

Figure 89: Catalogue for the 10mm diameter fasteners [25]

To connect the channel beams to the vertical racks, the fastener used can be found in Figure 90.

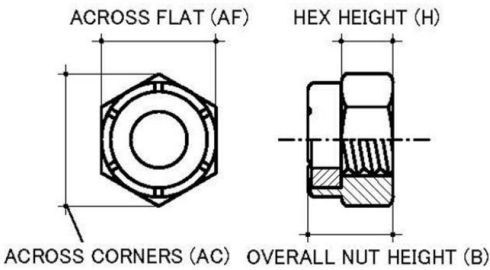
Technical Specifications



QFind	: 51S6M
System of Measurement	: Metric
Product	: Bolt Only
Head Type	: Hex Head
Size	: M6 (6mm)
Size / Diameter	: 6mm M6
Size & Thread Pitch	: M6 x 1.00p Coarse
Thread Pitch	: 1.00mm
Thread Direction	: Right Hand
Thread Type	: Metric - Coarse
Class / Grade	: A4-70 G316
Material	: Stainless Steel
Corrosion Protection Level	: Ultimate

Figure 90: Catalogue for the 6mm diameter fasteners. [26]

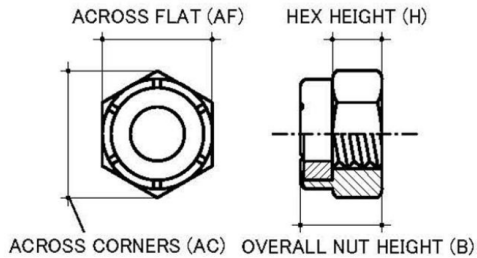
Technical Specifications



QFind	:	41NS10
System of Measurement	:	Metric
Product	:	Nut
Head Type	:	Hex Head
Nut Type	:	Nyloc Insert
Size	:	M10 (10mm)
Size / Diameter	:	10mm M10
Size & Thread Pitch	:	M10 x 1.50p Coarse
Thread Pitch	:	1.50mm
Thread Direction	:	Right Hand
Thread Type	:	Metric - Coarse
Class / Grade	:	A4-70 G316
Material	:	Stainless Steel
Corrosion Protection Level	:	Ultimate
Across Flats (af)	:	16.73 - 17.00mm
Across Corners (ac)	:	18.90 - 19.63mm
Spanner Size	:	17mm
Nut Height Overall (b)	:	9.64 - 10.00mm
Hex Height (h)	:	6.50mm
Closest size in imperial	:	3/8"
Standard Made To	:	DIN985

Figure 91: Catalogue for M10 Nylocs [27]

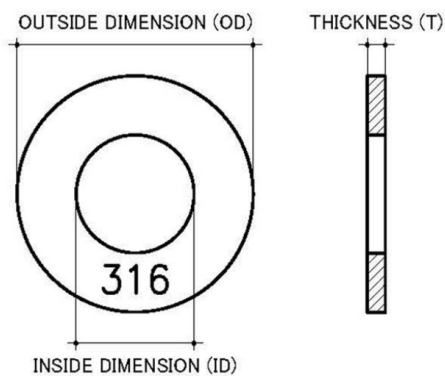
Technical Specifications



QFind	: 41NS6
System of Measurement	: Metric
Product	: Nut
Head Type	: Hex Head
Nut Type	: Nyloc Insert
Size	: M6 (6mm)
Size / Diameter	: 6mm M6
Size & Thread Pitch	: M6 x 1.00p Coarse
Thread Pitch	: 1.00mm
Thread Direction	: Right Hand
Thread Type	: Metric - Coarse
Class / Grade	: A4-70 G316
Material	: Stainless Steel
Corrosion Protection Level	: Ultimate
Across Flats (af)	: 9.78 - 10.00mm
Across Corners (ac)	: 11.05 - 11.50mm
Spanner Size	: 10mm
Nut Height Overall (b)	: 5.70 - 6.00mm
Hex Height (h)	: 4.00mm
Closest size in imperial	: 1/4"
Standard Made To	: DIN985

Figure 92: Catalogue for the M6 Nylocs [28]

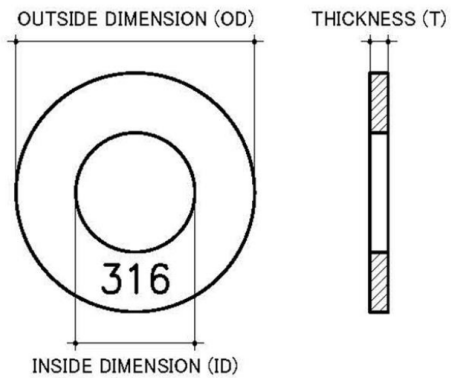
Technical Specifications



QFind	: 71S1021
System of Measurement	: Metric
Product	: Washer
Washer Type	: Standard
Size / Diameter	: 10mm M10
Washer Size	: M10
Class / Grade	: A4-70 G316
Material	: Stainless Steel
Corrosion Protection Level	: Ultimate
To Suit Fastener Size	: M10 3/8"
Inside Dimension (id)	: 10.20 - 10.75mm
Outside Dimension (od)	: 21.00mm
Thickness (t)	: 1.30mm
Standard Made To	: HEC Standard

Figure 93: catalogue for M10 washers to be used on either side of tower and beam connection. [29]

Technical Specifications



QFind	: 71S612
System of Measurement	: Metric
Product	: Washer
Washer Type	: Standard
Size / Diameter	: 6mm M6
Washer Size	: M6
Class / Grade	: A4-70 G316
Material	: Stainless Steel
Corrosion Protection Level	: Ultimate
To Suit Fastener Size	: M6 1/4" 12g 14g #12
Inside Dimension (id)	: 6.10 - 6.70mm
Outside Dimension (od)	: 12.50mm
Thickness (t)	: 1.30mm
Standard Made To	: HEC Standard

Figure 94: catalogue for M6 washers to be used beam side only. [30]

2.5 PARALLEL FLANGE CHANNELS

2

 Section: Parallel Flange Channels (PFC) Standard: AS/NZS 3679.1 Grade: 300MPa Surface Finish: Mill Finish							
Size Specification (mm)	Weight (kg/m)	Metre/Tonne	Depth of Section	Flange (mm)		Web (mm)	
				width	Thickness	Depth/Flange	Thickness
50 x 25*	3.83	261.1	50	25	—	—	—
75 x 40	5.92	169.0	75	40	6.1	62.8	3.8
100 x 50	8.33	120.0	100	50	6.7	86.6	4.2
125 x 65	11.90	84.0	125	65	7.5	110.0	4.7
150 x 75	17.70	56.5	150	75	9.5	131.0	6.0
180 x 75	20.90	47.6	180	75	11.0	158.0	6.0
200 x 75	22.90	43.7	200	75	12.0	176.0	6.0
230 x 75	25.10	39.8	230	75	12.0	206.0	6.5
250 x 90	35.50	28.2	250	90	15.0	220.0	8.0
300 x 90	40.10	24.9	300	90	16.0	268.0	8.0
380 x 100	55.20	18.1	380	100	17.5	345.0	10.0

2.5 Parallel Flange Channels

PARALLEL FLANGE CHANNELS LENGTH MATRIX					
Metric Designation (mm)	9.0m	10.5m	12m	13.5m	15m
75 x 40	✓	—	✓	—	—
100 x 50	✓	—	✓	—	—
125 x 65	✓	—	✓	—	—
150 x 75	✓	✓	✓	✓	✓
180 x 75	✓	✓	✓	✓	✓
200 x 75	✓	✓	✓	✓	✓
230 x 75	✓	✓	✓	✓	✓
250 x 90	✓	✓	✓	✓	✓
300 x 90	✓	✓	✓	✓	✓
380 x 100	✓	✓	✓	✓	✓

NOTE:

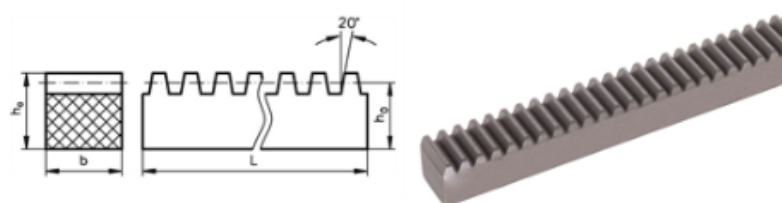
- » *Taper Flange Channel is available subject to enquiry.
- » Not all products are available at all locations. Please contact your Orrcon Steel / Metalcorp representative for availability.

40
E&OE



Figure 95: Catalogue page for the steel beams between the vertical racks and the tower wall. [31]

Gear rack made of steel module 8 tooth width 80mm height 80mm length 1500mm



Material: Specially treated bright steel C45KG. Milled teeth. Precisely Straightened. Cross-section tolerance $h_{11} = -1/10$ to $-2/10$ depending on size. Straight teeth. Pressure angle 20° . Tooth quality 8d25 modelled on DIN 3962, 3967, 3968. The teeth on the gear racks are manufactured using an overhead milling cutter. This leads to negative tolerances for dimension a and h_0 : up to Module 2: $-0,2$ mm. Module 2,5 to 3: $-0,3$ mm. Module 5 to 8: $-0,4$ mm. From Module 2, except for nominal length 500 mm, cut for continuous linking. * Length = Nominal length. The real length is roughly one multiple of the pitch.

Service: [Catalog Page](#) [Additional Documents](#)

The supplied 3D models, pictures and technical drawings are made with reasonable care. Nevertheless liability is excluded for the accuracy and correctness of this data.

Article ID: 24860300

Module	8
b [mm]	80
h_a [mm]	80
Nominal Length L [mm]	1500
Finish Size [mm]	1507,0 - 1,5
h_0 [mm]	72,0
Material	C45KG
Weight [kg]	66
Remark	

Figure 96: Catalogue page for the vertical racks. [32]